Supplementary material for

Building a Bilingual English–Spanish Parallel Corpus Using Multi-Model LLM Translation with Consensus Review

Mario Augusto Ramírez-Morales Esteban Castillo David Camilo Corrales
Victor Adrian Sosa-Hernández

This document has two parts. Section S1 contains the per-dataset measures that the article summarizes in its text, and Section S2 gives the configuration of the production run: implementation and hardware, the topic categories used for sampling, the parameters, and the prompts. The corpus, the code, and the notebooks are available at https://github.com/vsosahdz/SpanishDBText.

S1 Corpus measures

All measures were computed separately for English and Spanish and for each dataset (WikiText, FEVER, and Europarl), on the 12,003 entries of the released corpus.

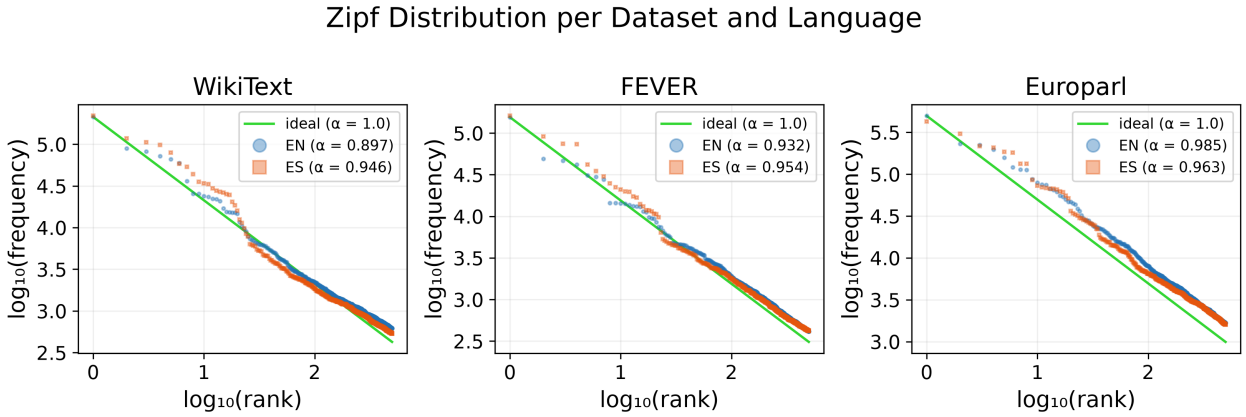


Figure S1: Zipf distribution (log-log) per dataset and language. The green line shows the ideal Zipf distribution ($\alpha = 1.0$) as a visual reference. Points above the line are words that appear more often than the ideal predicts (typically function words), and points below it are rare words in the long tail. Both languages follow near-linear trends close to the reference; the fitted exponents (0.89–0.99) are given in the word-frequency table of the article.

Table S1: General corpus statistics per dataset and language: total words, unique words, and words per document. Spanish has 17–57% more unique words than English in all three datasets.

Dataset	Language	Words	Unique	Words per document			
				Mean	Median	Min	Max
WikiText	EN	2,855,853	84,029	2,856	2,247	807	8,571
WikiText	ES	3,038,960	104,670	3,039	2,409	872	9,438
FEVER	EN	1,753,843	54,057	175	140	76	3,554
FEVER	ES	1,941,480	63,253	194	156	71	3,701
Europarl	EN	6,950,546	36,790	6,916	5,886	217	25,619
Europarl	ES	6,954,048	57,701	6,920	5,898	199	26,145

Table S2: Lexical diversity per dataset and language, computed at corpus level. TTR is the type-token ratio, which depends on text length; MATTR (window of 100 words) and MTLD do not. MTLD/doc mean is MTLD averaged over documents.

Dataset	Language	TTR	MATTR (w=100)	MTLD	MTLD/doc mean
WikiText	EN	0.029	0.708	84.63	84.69
WikiText	ES	0.034	0.695	77.27	77.83
FEVER	EN	0.031	0.690	72.86	68.44
FEVER	ES	0.033	0.670	63.24	61.19
Europarl	EN	0.005	0.715	89.74	89.29
Europarl	ES	0.008	0.717	91.47	91.54

Table S3: Sentence length per dataset and language, in words. The maximum values are artifacts of the regular-expression sentence splitter on text without standard punctuation, not real sentences.

Dataset	Language	Sentence length (words)			
		Mean	Median	Standard deviation	Maximum
WikiText	EN	22.0	21	11.9	403
WikiText	ES	23.8	22	12.9	501
FEVER	EN	21.4	20	12.8	469
FEVER	ES	24.4	22	14.4	658
Europarl	EN	25.0	22	15.2	1,584
Europarl	ES	24.9	22	15.2	1,616

Table S4: Quality indicators per dataset and language: character expansion ratio and its standard deviation, exact duplicates within each dataset, and the shares of stopwords and special characters.

Dataset	Language	Expansion	Exp. std	Duplicates	Stopwords %	Special %
WikiText	EN	N/A	N/A	0	36.67	3.08
	ES	1.044	0.036	0	38.71	2.47
FEVER	EN	N/A	N/A	11	33.16	3.24
	ES	1.091	0.066	0	38.89	2.44
Europarl	EN	N/A	N/A	0	43.16	3.23
	ES	1.051	0.021	0	40.68	3.05

Expansion ratio = character length of Spanish / English (shown on the ES row). Duplicates are exact-match counts within each dataset.

Table S5: Vocabulary growth as datasets are processed sequentially, per language. Bold marks the Europarl row, which adds the most words of the three datasets but the fewest new ones, and where English and Spanish diverge most. The 100% values for WikiText are definitional, since it is processed first.

Dataset	Language	Words Added	Unique After	New Unique	New (%)
WikiText	EN	2,855,853	84,029	84,029	100.0
	ES	3,038,960	104,670	104,670	100.0
+ FEVER	EN	1,753,843	107,163	23,134	21.6
	ES	1,941,480	129,623	24,953	19.3
+ Europarl	EN	**6,950,546**	117,297	**10,134**	**8.6**
	ES	**6,954,048**	150,745	**21,122**	**14.0**

S2 Pipeline configuration

This section gives the configuration of the production run that the article summarizes in its Implementation and Pipeline Configuration sections. The code and the full text of every prompt are in the repository.

S2.1 Implementation and hardware

The pipeline is written in Python 3.11+ and uses no NLP or translation libraries. API calls are made with httpx, and the translation models run in parallel with Python's ThreadPoolExecutor. Model assignments, provider credentials, and generation parameters (temperature, context window, and output limit) are read from a `.env` file; the other thresholds and the retry counts are constants in the pipeline code. Subset selection and corpus analysis were done in Jupyter notebooks with pandas and NumPy, and all outputs are stored as JSON files.

We ran the whole production from one laptop: an 8-core AMD Ryzen 9 8945HS with 32 GB of RAM and an NVIDIA RTX 4060 Laptop GPU with 8 GB of VRAM, under Windows 11. The same machine served Qwen 3.5 9B with Ollama inside a Docker container. The other five model roles were reached through Azure OpenAI and Azure AI Foundry.

S2.2 Topic categories for stratified sampling

WikiText and FEVER, both taken from Wikipedia, share 14 categories: music, military history, sports, film and television, geography and places, science, transport, biography, politics and government, nature and weather, technology, history, religion and philosophy, and *other*. Europarl uses 15 categories adapted to parliamentary business: economy and finance, institutional affairs, foreign affairs, trade and market, environment and energy, human rights, transport and infrastructure, development aid, agriculture and food, justice and security, social affairs and employment, health and consumers, digital technology, defence, and *other*.

Each entry goes to the category whose keyword list has the most matches in the entry, or to *other* when no keyword matches. The text matched against the keyword lists differs by source. For WikiText and FEVER, it is the title plus the first 2,000 characters of the body; for Europarl, it is the debate title alone, because the title of a session already states its subject.

S2.3 Parameters

Table S6 lists the parameters of the run. We used a temperature of 0.2 for all models, so that translations are mostly stable but can vary slightly. A value of 0 would remove all variation, and higher values would add randomness to a task that requires fidelity to the source. The context window (32,768 tokens) and the maximum output (8,192 tokens for translation and review, 1,024 for evaluation and metadata) are the limits supported by the models used. The article explains the chunking values in its Pipeline Configuration section.

We set the API timeout at 600 seconds (10 minutes) for the local Qwen model, which is much slower than the cloud models on long chunks. The override margin of 10 points means that the programmatic truncation check replaces the completeness score of the evaluator only when the two differ widely, which avoids corrections for small differences. The 40% weight of completeness in the overall score reflects the priority given to missing content: truncation is a more serious problem than minor issues of fluency or clarity.

Table S6: Key pipeline parameters.

Parameter	Scope	Value
Temperature	All models	0.2
Context window	All models	32,768 tokens
Max output	Translation, Review	8,192 tokens
Max output	Evaluation, Metadata	1,024 tokens
Chunking threshold	Stage 2	6,000 est. tokens
Max chunk size	Stage 2	12,000 characters
API timeout	All calls	600 seconds
Max retries	All calls	3
Language retries	Stage 3	2
Truncation ratio threshold	Stage 5	0.92
Truncation override margin	Stage 5	10 points
Completeness weight	Stage 5	40%

S2.4 Prompts

Every LLM call has a *system prompt*, which defines the model's role and fixed rules, and a *user prompt*, which provides the text of each entry and any extra context. Table S7 lists all the prompts of the pipeline.

Table S7: Prompt inventory: structural components and corpus-specific variants.

Prompt	System layer	User layer	Output	Variants
Translation system (St. 3)	Role + 10 fidelity/style rules	Source text + direction	Plain text	3 (WT, F, EP)
Language retry user (St. 3)	(same as translation)	Error notice + source text	Plain text	1 per direction
Critique system + user (St. 4a)	Role + JSON schema requirement	Orig. + candidates	JSON	1 (shared)
Merge system + user (St. 4b)	Role + fidelity/completeness rules	Orig. + candidates + critique	Plain text	1 (shared)
Evaluation system + user (St. 5)	Role + scoring guide + completeness rules	Orig. + translation + char lengths + ratio	JSON	1 (shared)
Metadata system + user (St. 6)	Role + format instruction	4-field template + article text	Structured lines	2 (EN, ES)

Translation. The translation system prompt gives the model a professional role ("You are a professional translator specialized in editorial and technical translation") and sets ten rules. They cover semantic fidelity (no invention, omission, or addition), natural output, a neutral professional Spanish register not tied to any national variety, preservation of technical terms and proper nouns, respect for the original tone and paragraph structure, and a ban on markdown formatting that is not in the original.

There are three variants of this system prompt, one per corpus, which keep the same role and core rules and adapt them to the structure of each corpus. The WikiText variant adds rules to remove the `=` signs of section headers and the `<unk>` tokens of the dataset. The FEVER variant adds a sports terminology rule: use the established Spanish term when one exists (e.g., *tiro libre* for *free throw*, *ponchar* for *strikeout*), and keep the English term only when none exists. The Europarl variant replaces the WikiText cleaning rules with an instruction to keep all XML structural tags (`<CHAPTER>`, `<SPEAKER>`, `<P>`) intact and untranslated.

The user prompt is a short wrapper that provides the text and states the translation direction. When language validation finds output in the wrong language, the retry user prompt reports the error (e.g., "Your previous response was in English. You MUST translate the following text to Spanish") and repeats the original text.

The system prompts for translation, review, and evaluation are written in Spanish, to orient the model toward the target language. Only the metadata stage has one system prompt per language, one in Spanish and one in English, because it extracts metadata from both the source and the translation.

Review. The critique system prompt asks the reviewer to analyze each candidate, list its strengths and weaknesses, and give it a score from 0 to 100. It must return a JSON object that also names the base candidate and the improvements to apply. The critique user prompt supplies the original text and the candidates in clearly marked sections. The merge system prompt then asks the reviewer to write the final translation from the original text, the candidates, and the critique. Its rules repeat the fidelity and completeness rules of the translation prompt, and its output is plain text. The critique runs once per entry, and its choice of base candidate applies to the whole entry. The merge runs once per segment for entries that were split, at the same boundaries as the chunks.

Evaluation. The evaluation system prompt asks for strict scoring (a perfect score must be extremely rare) and contains the scoring guide of the article. It also sets completeness rules: if the translation is much shorter than the original (character ratio below 0.85), it is probably truncated, and a truncated translation cannot receive more than 70 for completeness. The user prompt gives the character counts of both texts and their ratio, and adds a warning when the ratio is below 0.85. The output is a JSON object with the four dimension scores, a list of errors, and an overall score.

Metadata. The metadata system prompt combines a minimal role with a strict format requirement. It has two variants, one in Spanish for the translation and one in English for the source, so that the model works in the language of the text it analyzes. The user prompt is a fixed template with one line per field (title, keywords, topic, and summary), format limits for each field, and the text in a separate, clearly marked block. The pipeline parses the answer with regular expressions. Each corpus has its own topic list (Table S8), and all three lists end with an open option ("or other appropriate").

Table S8: Topic categories offered to the metadata model in Stage 6, per corpus. Each list ends with an open-ended "or other appropriate" option.

Corpus	Categories
WikiText-103	video game, novel, biography, science, history, technology, politics, medicine, sports, entertainment
FEVER	sports, music, film, history, politics, science, geography, biography, technology, religion, military, transport
Europarl	politics, legislation, economy, environment, human rights, agriculture, defense, trade, health, technology